

The Hodge Complex on the 8×8 Torus: Differential Forms, Boson Fermion Pairing, and the Correct Formulation of the SYK Substrate

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Abstract

The discrete substrate of the WIN Paradigm has been modeled as a lattice of Majorana fermions with one fermion per site. This paper establishes that the correct formulation is the *Hodge complex* of differential forms on the 8×8 torus. The operator $D = d + d^*$ satisfies $D^2 = \Delta$ exactly on the discrete torus, without fermion doubling, and provides a canonical $\mathbb{Z}_2$ -graded structure: even forms ($\Omega^0 \oplus \Omega^2$) carry bosonic statistics, odd forms (Ω^1) carry fermionic statistics. The scalar Laplacian's 14-fold degeneracy at $\lambda = 4$ lifts to every form degree, appearing as 14 scalar, 28 vector, and 14 pseudo-scalar modes, with exact boson fermion multiplicity matching at every eigenvalue. The spectrum is supersymmetric with D as the supercharge. The Ω^0 zero mode acquires a 1-loop mass $m_{\text{eff}}^2 = 4.55 g$ on the physical torus, with logarithmic growth in system size that bounds the bosonic sector to the UV and leaves the fermionic sector to dominate the infrared. The single-kernel distribution of the geometric D_4 -invariant kernel is non-Gaussian (KS $p < 10^{-200}$), but sums over many independent contributions converge to Gaussian by the central limit theorem (KS $p = 0.048$ at 50 terms). We state this precisely as a *marginal* CLT limit and identify the joint-independence question as an open problem for the melonic structure of the geometric ensemble. Monte Carlo analysis of the 14-mode multiplet confirms the binary boson/fermion split, the ternary D_4 -orbit split, and the N/4 mass ordering (Spearman $\rho = -0.9965$), while establishing that the finer Standard Model structure specific charge assignments and generation organization is not derivable from the mode coordinates alone and is empirical input. We catalogue six corrections required in previously circulated WIN papers, and list four open problems.

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1 Introduction

The microscopic layer of the WIN Paradigm has been modeled as a Sachdev-Ye-Kitaev (SYK) ensemble of N Majorana fermions placed one per site on the 8×8 torus [7, 8, 9]. This modeling choice has three structural problems:

1. Nothing in the torus geometry specifies that physical degrees of freedom should sit on vertices. The torus is a spectral object; the fermions are imposed on it.
2. The naive spinorial Dirac operator on the lattice suffers from fermion doubling: D_{spin}^2 is a doubled Laplacian, not the nearest-neighbor Laplacian that produces the 14-fold degeneracy at $\lambda = 4$.
3. The Gaussian SYK ensemble is assumed. No mechanism is given for why the couplings should be Gaussian random variables.

This paper corrects (1) and (2) by reformulating the substrate as a *Hodge complex*, and partially addresses (3) by showing that a central-limit-theorem (CLT) argument connects the geometric kernel to the Gaussian SYK ensemble at the level of marginal distributions. We are explicit about what is not resolved: the joint independence structure required for melonic dominance, the rigorous RG flow to the SYK fixed point, and the fine structure of particle identification are all left as open problems.

Scope. This paper establishes the correct mathematical substrate of the WIN Paradigm. It does not claim to derive holography, does not claim to derive the Standard Model charge assignments, and does not claim that the CLT argument is complete. Where the paper reaches a limit, that limit is stated.

Organization. Section 2 introduces the Hodge–Dirac operator. Section 3 derives its spectrum and the lift of the 14-fold degeneracy. Section 4 quantizes the complex and derives boson–fermion statistics. Section 5 discusses the supercharge structure. Section 6 derives the geometric quartic interaction. Section 7 computes the boson gap. Section 8 discusses the infrared behavior. Section 9 presents the marginal CLT result. Section 10 addresses particle identification. Section 11 catalogues corrections to prior work. Section 12 lists open problems. Section 13 concludes.

2 The Hodge Complex on the 8×8 Torus

2.1 Differential forms

Let $\Lambda = \mathbb{Z}_8 \times \mathbb{Z}_8$ be the periodic square lattice with $N = 64$ sites. The Hodge complex consists of:

- Ω^0 : 0-forms (scalar functions), 64-dimensional.
- Ω^1 : 1-forms (edge fields), 128-dimensional (two components per site).
- Ω^2 : 2-forms (plaquette fields), 64-dimensional.

The exterior derivative d and codifferential d^* are the standard lattice operators. In local coordinates with e_1, e_2 the lattice basis vectors:

$$(d_0 f)(x, \mu) = f(x + e_\mu) - f(x), \tag{1}$$

$$(d_1 \omega)(x) = \omega_1(x) + \omega_2(x + e_1) - \omega_1(x + e_2) - \omega_2(x), \tag{2}$$

$$(d_0^* \omega)(x) = \sum_{\mu} [\omega_{\mu}(x) - \omega_{\mu}(x - e_{\mu})], \tag{3}$$

$$(d_1^* \chi)(x, \mu) = \pm [\chi(x) - \chi(x + e_{\mu})], \tag{4}$$

with the sign chosen so that $d_0^* d_1^* = 0$. They satisfy $d_1 d_0 = 0$ exactly.

2.2 Hodge-Dirac

Definition 1 (Hodge-Dirac operator). The Hodge-Dirac operator is

$$D = d + d^*, \quad (5)$$

acting on $\Omega^0 \oplus \Omega^1 \oplus \Omega^2$, a total of $64 + 128 + 64 = 256$ real degrees of freedom.

Theorem 2 (Hodge-Dirac square). *On the discrete torus,*

$$D^2 = \Delta \cdot \mathbb{K}_{\Omega^k} \quad \text{on each form degree } k, \quad (6)$$

where Δ is the scalar 5 point Laplacian

$$(\Delta f)(x) = 4f(x) - \sum_{|e|=1} f(x+e). \quad (7)$$

Proof. Direct computation. The nilpotency $d_1 d_0 = 0$ gives $\{d, d^*\} = 0$ on the diagonal blocks, and the anticommutator $\{d, d^*\}$ equals Δ on each form degree. $\square$

Remark 3. Unlike the spinorial Dirac operator on the same lattice, which suffers fermion doubling (D_{spin}^2 is a doubled Laplacian), the Hodge-Dirac square equals the scalar Laplacian exactly. This is the structural reason we adopt the Hodge complex.

3 Spectrum and the Lift of the 14-Fold Degeneracy

3.1 Scalar Laplacian

The scalar Laplacian on the 8×8 torus has the spectrum

$$\lambda(n_1, n_2) = 4 - 2 \cos(2\pi n_1/8) - 2 \cos(2\pi n_2/8). \quad (8)$$

The 13 distinct eigenvalues and their multiplicities are shown in Table 1.

Table 1: Scalar Laplacian spectrum on the 8×8 torus and its lift to the Hodge complex. The 14-fold degeneracy at $\lambda = 4$ is the defining feature of the FDLM/WIN substrate.

λ	$m(\Omega^0)$	$m(\Omega^1)$	$m(\Omega^2)$
0.0000	1	2	1
0.5858	4	8	4
1.1716	4	8	4
2.0000	4	8	4
2.5858	8	16	8
3.4142	4	8	4
4.0000	14	28	14
4.5858	4	8	4
5.4142	8	16	8
6.0000	4	8	4
6.8284	4	8	4
7.4142	4	8	4
8.0000	1	2	1
Total	64	128	64

3.2 Lift to the Hodge complex

Theorem 4 (14-fold lift). *The 14-fold degeneracy of the scalar Laplacian at $\lambda = 4$ lifts to the Hodge complex as*

$$14 \text{ modes in } \Omega^0, \quad 28 \text{ modes in } \Omega^1, \quad 14 \text{ modes in } \Omega^2. \quad (9)$$

Proof. Since $D^2 = \Delta$ on each form degree, the spectrum of D^2 acting on Ω^k is the scalar Laplacian eigenvalues with the indicated multiplicities. At $\lambda = 4$, the multiplicity is 14 in Ω^0 , 28 in Ω^1 (two components per site, each contributing 14), and 14 in Ω^2 . $\square$

The 28-fold degeneracy in Ω^1 is the geometric origin of the boson-fermion pairing.

Remark 5 (Why 8×8). On an $L \times L$ torus, the midpoint multiplicity at the self-paired eigenvalue is exactly $2(L-1)$. Setting this equal to 14 gives $L = 8$. The torus size is therefore derived from the Standard Model particle count, not chosen. Larger tori would have different multiplicities and do not host the 14-mode multiplet.

4 Quantization and Statistics

4.1 The $\mathbb{Z}_2$ grading

Definition 6 (Parity operator). Define the parity operator $(-1)^F$ by

$$(-1)^F = \begin{cases} +1 & \text{on } \Omega^0 \oplus \Omega^2, \\ -1 & \text{on } \Omega^1. \end{cases} \quad (10)$$

The Hodge–Dirac is odd: $D(-1)^F = -(-1)^F D$.

4.2 Field quantization

Assign canonical commutation relations to even-sector modes a_n and canonical anticommutation relations to odd-sector modes ψ_n :

$$[a_n, a_m^\dagger] = \delta_{nm}, \quad \{\psi_n, \psi_m^\dagger\} = \delta_{nm}. \quad (11)$$

The Hodge–Dirac becomes the supercharge

$$D = \sum_n \sqrt{\lambda_n} (a_n^\dagger \psi_n + \psi_n^\dagger a_n). \quad (12)$$

4.3 Statistics

Theorem 7 (Form-degree statistics). *The three sectors of the Hodge complex carry the following statistics:*

- Ω^0 (0-forms): bosonic.
- Ω^1 (1-forms): fermionic.
- Ω^2 (2-forms): bosonic.

Proof. The $\mathbb{Z}_2$ grading $(-1)^F$ defines bosonic (even) and fermionic (odd) sectors. Field quantization with commutators on the even sector and anticommutators on the odd sector gives the stated statistics. $\square$

Fermionic statistics has a geometric origin: it is the $\mathbb{Z}_2$ -odd part of the Hodge complex.

5 The Supercharge Structure

The superalgebra is

$$\{Q, Q^\dagger\} = 2H, \quad Q^2 = (Q^\dagger)^2 = 0, \quad (13)$$

with $Q = D$ and $H = D^2/2 = \Delta/2$. This is the standard $\mathcal{N} = 1$ supersymmetric quantum mechanics algebra.

Every bosonic mode at energy λ_n has a fermionic partner at the same energy. The spectrum is supersymmetric at every eigenvalue, including the 14-fold at $\lambda = 4$:

$$28 \text{ bosons} + 28 \text{ fermions at } \lambda = 4. \quad (14)$$

The exact equality of bosonic and fermionic multiplicities at every eigenvalue is the statement that the Hodge complex realizes an $\mathcal{N} = 1$ supersymmetric quantum mechanics on the 8×8 torus.

6 The Geometric Quartic Interaction

6.1 D_4 -invariant quartics

In local coordinates, at each site the 1-form has two components (ω_1, ω_2) corresponding to the e_1 and e_2 lattice directions. The D_4 -invariant quartic polynomials are generated by two independent terms:

$$V_A(x) = \omega_1(x)^4 + \omega_2(x)^4, \quad V_B(x) = \omega_1(x)^2 \omega_2(x)^2. \quad (15)$$

The most general D_4 -invariant quartic is $g_A V_A + g_B V_B$ with two real couplings.

Remark 8. The invariance group is $D_4 = \langle r, s \mid r^4 = s^2 = (rs)^2 = e \rangle$ acting on the pair (ω_1, ω_2) by $r : (\omega_1, \omega_2) \mapsto (-\omega_2, \omega_1)$ and $s : (\omega_1, \omega_2) \mapsto (\omega_1, -\omega_2)$. Direct check: V_A, V_B are the only D_4 -invariant quartics up to linear combination.

6.2 The Hodge Hamiltonian

The full Hamiltonian is

$$H_{\text{Hodge}} = \frac{1}{2} D^2 + \sum_x (g_A V_A(x) + g_B V_B(x)). \quad (16)$$

The first term is the free supercharge Hamiltonian; the second is the interaction.

6.3 Comparison with SYK

The SYK Hamiltonian

$$H_{\text{SYK}} = \sum_{i < j < k < l} J_{ijkl} \psi_i \psi_j \psi_k \psi_l \quad (17)$$

with Gaussian random J is *not* equal to H_{Hodge} . As discussed in Section 9, it is an approximation at the level of marginal distributions.

7 The Boson Gap

7.1 One-loop mass of the Ω^0 zero mode

The Ω^0 zero mode (the constant function) has $\lambda = 0$ in the free theory. At one loop, the quartic interaction generates a mass:

$$m_{\text{eff}}^2 = 12g \langle \phi^2 \rangle_{\text{free}}, \quad (18)$$

where

$$\langle \phi^2 \rangle = \frac{1}{N} \sum_{\lambda_n \neq 0} \frac{1}{\lambda_n}. \quad (19)$$

7.2 Numerical evaluation

Table 2: Boson zero-mode 1-loop mass squared as a function of torus size. The scaling is logarithmic: $\langle \phi^2 \rangle \sim (1/2\pi) \ln L$.

L	$N = L^2$	$\langle \phi^2 \rangle$	m_{eff}^2/g
4	16	0.268	3.22
6	36	0.333	4.00
8	64	0.379	4.55
10	100	0.415	4.98
12	144	0.444	5.33
16	256	0.490	5.88

The measured slope of $\langle \phi^2 \rangle$ with respect to $\ln L$ is 0.160, matching the analytic $1/(2\pi) = 0.159$ to three digits.

7.3 Physical consequence

On the physical 8×8 torus:

$$m_{\text{eff}}^2 = 4.55g, \quad m_{\text{eff}} = 2.13\sqrt{g}. \quad (20)$$

The boson zero mode is gapped. The gap grows logarithmically with system size. In the thermodynamic limit, the bosons become infinitely massive and decouple from the IR.

The boson sector is a UV effect. The fermion sector dominates the IR.

8 Infrared Behavior

The fermionic sector Ω^1 has no mass gap. Its low-energy dynamics is *consistent with* standard SYK: maximal chaos, MSS saturation, and flow to the 0+1-dimensional conformal fixed point with scaling dimension $\Delta = 1/4$.

The bosonic sector $\Omega^0 \oplus \Omega^2$ is gapped with $m_{\text{eff}} \sim 2.13\sqrt{g}$. At energies $E \ll m_{\text{eff}}$, the bosons are frozen out.

Proposition 9 (Boson decoupling). *On the physical torus, the boson sector decouples from the IR. The low-energy behavior of the Hodge complex is determined by the fermionic sector Ω^1 .*

Remark 10 (Scope of the holographic claim). The fermionic sector is consistent with SYK, and SYK has a known JT dual in nearly-AdS₂. We do *not* claim that the discrete Hodge complex on a finite torus rigorously flows to a gravitational dual. The chain “discrete Hodge $\rightarrow$ SYK $\rightarrow$ JT” has a gap between the first two arrows: a rigorous RG analysis of the discrete-to-continuum limit is required and is left as an open problem (Section 12). What the present paper establishes is the structural compatibility.

9 The Marginal CLT Limit of the SYK Ensemble

9.1 Statistical test of the geometric kernel

The natural D_4 -invariant geometric kernel on the torus is

$$K(i, j, k, l) = d(i, j) + d(k, l), \quad (21)$$

where d is the toroidal distance. From 2×10^5 samples of 4-tuples on the 8×8 torus:

- Mean: 6.286, standard deviation: 1.609.
- Skewness: -0.075 (Gaussian: 0).
- Kurtosis: 2.652 (Gaussian: 3).
- Kolmogorov–Smirnov test against Gaussian: $p < 10^{-200}$.

The single-kernel distribution is *not* Gaussian.

9.2 Marginal CLT convergence

The effective SYK coupling for a given mode is a sum over many independent geometric contributions. We test the CLT hypothesis by computing the distribution of a sum of independent kernel values:

Table 3: Convergence of the *marginal* distribution to Gaussianity as the number of independent contributions increases.

Distribution	Skewness	Kurtosis	KS p vs Gaussian
Single kernel	-0.075	2.652	$< 10^{-200}$
4-pair cycle	-0.034	3.082	2×10^{-74}
6-pair sum (squared)	-0.414	2.837	$< 10^{-200}$
50-term sum	$+0.061$	3.012	0.048

The 50-term sum is Gaussian at the level of its marginal distribution: skewness $+0.061$, kurtosis 3.012, KS $p = 0.048$.

9.3 What is and is not established

Theorem 11 (Marginal CLT limit). *The marginal distribution of a large sum of independent geometric kernel values converges to a Gaussian by the Lindeberg–Lévy central limit theorem. The convergence rate is $O(1/\sqrt{N_{\text{terms}}})$ from Berry–Esseen.*

Proof. The single-kernel distribution has finite variance $\sigma_K^2 = 1.61^2$, so the CLT applies. The Berry–Esseen bound gives $O(1/\sqrt{N_{\text{terms}}})$ convergence in the Kolmogorov metric. $\square$

Remark 12 (What this does *not* establish). The CLT result is about *marginal* distributions. It does not establish that the joint distribution of SYK couplings J_{ijkl} is independent across different index tuples. In standard SYK, melonic dominance requires that non-melonic diagrams vanish in the large- N limit, and this in turn requires that couplings on overlapping tuples are sufficiently uncorrelated.

The geometric kernel has correlations: overlapping tuples share sites, so $K(i, j, k, l)$ and $K(i, j, k, m)$ are correlated. Whether these correlations are sub-leading in $1/N$ — and therefore whether melonic dominance survives — is an open problem. We state this explicitly.

10 Particle Identification

10.1 Derived structures

The 14 patterns at $\lambda = 4$ have three derived structures:

(1) **Binary split.** The Hodge $\mathbb{Z}_2$ grading splits the 14 modes into 10 fermionic and 4 bosonic modes, matching the Standard Model content (10 fermions, 4 bosons per generation).

(2) **Ternary split.** The D_4 orbit structure partitions the 14 modes into $8 + 4 + 2$, matching the orbit sizes $\mathcal{O}_1, \mathcal{O}_2, \mathcal{O}_0$.

(3) **Mass ordering.** The $N/4$ label orders the 14 modes by observed mass with Spearman $\rho = -0.9965$, $p = 4 \times 10^{-12}$.

10.2 Monte Carlo verification

We verify these structures by k -means clustering on the feature space (support, $N/4$, orbit, form, n_1, n_2):

- $k = 2$: cluster sizes $[10, 4]$ — matches the fermion/boson split.
- $k = 3$: cluster sizes $[8, 4, 2]$ — matches the D_4 orbit structure.

Both structures are statistically significant.

10.3 Fine structure not derivable

The finer Standard Model structure — the partition into 3 charged leptons, 3 up-quarks, 3 down-quarks, 1 neutrino, and 4 bosons — is *not* derivable from the mode coordinates. Two independent tests confirm this:

Monte Carlo permutation test. The SM labeling is weakly special: $p = 0.045$. The D_4 orbit labeling is $3\times$ more natural in feature space ($F = 1.96$ vs 0.59 for SM).

Constructive algebraic test. No combination of the framework ratios $w_u = 3/10$, $w_d = 7/10$, $a = 3/5$, $\lambda = 14$, $\text{corr} = 20/17$ produces a clean 5-way partition. Threshold classification gives at most 81% agreement. Fisher discriminant analysis shows uniform scores of $F \approx 2.0$, a signature of no real structure.

10.4 No post-hoc tuning

The particle identification results are not the outcome of a parameter search. The $N/4$ values are derived from the δ -function rule (8 branches, each a polynomial in $d = 4$ and $H = 16$), and the mass map uses two parameters $(A, B) = ((2d - 1)N/2, (d - 1)kL^2/(2N^2))$, both derived from framework constants. No parameters were fit to the observed mass data. The derivation is testable: if the δ -function rule is modified, the $N/4$ values change and the correlation with observed masses degrades. Conversely, if the rule is correct, the $\rho = -0.9965$ correlation is a prediction, not a fit.

10.5 Electric charge

Every simple formula of $(n_1, n_2, \text{support})$ tested gives RMS charge error > 0.6 . Electric charge is not derivable from the mode coordinates alone.

10.6 Conclusion

Theorem 13 (Particle identification scope). *The torus geometry determines the coarse structure of the particle multiplet: the boson fermion split, the orbit decomposition, and the mass ordering. It does not determine the fine structure of the Standard Model charge assignments or generation organization. The fine structure is empirical input, on the same footing as the Yukawa couplings in the Standard Model.*

This is the same boundary as the Standard Model itself: the gauge structure and generation count are derived; the specific fermion masses and mixings are empirical.

11 Corrections to Prior Work

We catalogue six corrections that should be applied to previously circulated WIN papers.

11.1 Correction 1 - The site-fermion picture

Prior: SYK TENSOR papers place one Majorana fermion per torus site (64 Majoranas).

Corrected: The substrate is the Hodge complex. Fermions are odd-form quanta in Ω^1 , 128 real degrees of freedom (equivalent to 64 complex fermions).

11.2 Correction 2 - The dimension match

Prior: Implicit claim that the Hodge complex dimension (256) equals the SYK Hilbert space dimension at $N = 16$ (256).

Corrected: These are different counts. 256 is the number of real components of the Hodge complex. The fermionic Hilbert space is 2^{64} , not 2^8 .

11.3 Correction 3 - $G_{\text{eff}} = 1/\lambda_L$

Prior: SYK TENSOR papers claim the chaos-gravity correspondence is quantitative.

Corrected: In JT gravity, the Newton constant and the Lyapunov exponent are both set by the Schwarzian coupling $C = N/(4J)$, but they are not inverses of one another. $G_N \lambda_L$ is temperature-dependent at low T ($\sim JT/(4N)$) and coupling-dependent at high T ($\sim cJ^2/(8\pi N)$). The relation is qualitative, not quantitative.

11.4 Correction 4 - $\gamma \approx 1.0$ – 1.2

Prior: SYK TENSOR papers cite $\gamma \approx 1.0$ – 1.2 .

Corrected: The stability coefficient is $\gamma = 0.702839\dots$ from DME. The 1.0–1.2 value arose from a Gaussian variance error ($\sigma^2 = 1/4$ should be $1/2$) and an unjustified relation $\gamma = \sqrt{2|b|}$.

11.5 Correction 5 - 4D gravity from 2D SYK

Prior: SYK TENSOR papers claim 4D gravity emerges from scrambling.

Corrected: The 2D SYK array flows to 0+1-dimensional SYK in the IR. Its JT dual is AdS_2 , not AdS_4 . No propagating graviton. The 4D substrate is provided by FDLM/WIN.

11.6 Correction 6 - Architecture

Prior: SYK implied to produce the 4D vacuum.

Corrected: FDLM/WIN produces the $D = 4$ substrate. SYK stabilizes it via the Hodge Dirac and the antipode identity.

12 Open Problems

12.1 Joint independence and melonic structure

The CLT result of Section 9 is for the *marginal* distribution of the effective coupling. Standard SYK melonic dominance requires that the couplings J_{ijkl} are sufficiently independent across different index tuples so that non-melonic diagrams vanish in the large- N limit. The geometric kernel $K(i, j, k, l) = d(i, j) + d(k, l)$ has correlations between overlapping tuples (e.g., tuples differing in one index share sites). Whether these correlations are sub-leading in $1/N$ is not established. This is the most important open problem in the paper.

What would close it: A direct calculation of the joint correlation function $\langle K(i, j, k, l) K(i, j, k, m) \rangle$ and its scaling with N . If correlations scale as $O(1/N)$ relative to the diagonal, melonic dominance survives; if they scale as $O(1)$, the geometric ensemble is not SYK-like.

12.2 Rigorous RG flow to the SYK fixed point

Section 8 states that the IR of the fermionic sector is *consistent with* SYK. A rigorous derivation of the flow from the discrete Hodge complex to the SYK fixed point requires an analytic renormalization-group analysis of the discrete-to-continuum limit, including controlled bounds on irrelevant operators. This is not attempted here.

What would close it: A Wilsonian RG calculation of the effective action for the Ω^1 sector, showing that the only relevant operators in the IR are those of the SYK universality class.

12.3 Extension of the CLT argument to joint distributions

The 50-term sum test establishes convergence of the marginal distribution. The joint distribution of the full SYK ensemble requires a multivariate CLT argument with explicit treatment of the covariance structure. This is a generalization of Section 9 that has not been done.

What would close it: A Berry–Esseen-type bound on the multivariate distribution of the SYK coupling matrix, given the covariance structure of the geometric kernel.

12.4 Fine structure of particle identification

The partition of the 14 modes into 3 charged leptons, 3 up-quarks, 3 down-quarks, 1 neutrino, and 4 bosons is not derivable from the mode coordinates. Whether it is derivable from an extension of the framework (e.g., via the CKM/PMNS sector, or via additional geometric invariants) is an open question. The present paper establishes only the coarse structure (binary and ternary splits, mass ordering).

What would close it: A constructive derivation of the 5-way SM partition from framework constants, analogous to the CKM derivation using $w_u, w_d, a, \lambda, \text{corr.}$ If no such derivation exists, the fine structure is empirical input, and the paper’s Theorem 7-scope statement is the final answer.

13 Conclusion

The discrete substrate of the WIN Paradigm is the Hodge complex of differential forms on the 8×8 torus. The operator $D = d + d^*$ is the supercharge; $D^2 = \Delta$ exactly; the spectrum is supersymmetric; the 14-fold degeneracy at $\lambda = 4$ lifts to 28 in the fermionic sector.

The bosons in Ω^0 are gapped by a 1-loop quartic interaction with $m_{\text{eff}}^2 = 4.55g$ on the physical torus. In the thermodynamic limit, the bosons decouple. The IR is dominated by the fermionic sector, which is *consistent with* 0+1-dimensional SYK.

The single-kernel geometric distribution is non-Gaussian, but sums over many independent contributions converge to Gaussian by the central limit theorem. We state this as a *marginal*

CLT result, and identify the joint-independence question as the key open problem for melonic structure.

The particle identification is bounded: the binary boson/fermion split, the ternary orbit split, and the $N/4$ mass ordering are derived; the fine structure (charge assignments, generation organization) is empirical.

Six corrections are catalogued for prior WIN papers. Four open problems are stated.

The Hodge complex is the correct formulation of the substrate. The site-fermion picture is replaced by the form-quantum picture. The substrate is a pressurized toroidal grid of coupled 0-, 1-, and 2-forms, and the fermions are its odd-degree excitations.

A Numerical Summary

Table 4: Complete numerical values.

Quantity	Value
Torus size	8×8
Hodge complex dimension	256
Ω^0 dimension	64
Ω^1 dimension	128
Ω^2 dimension	64
14-fold degeneracy	$\lambda = 4$ in Ω^0, Ω^2
28-fold degeneracy	$\lambda = 4$ in Ω^1
$\dim \ker D$	$4 = b_0 + b_1 + b_2$
Boson gap at $L = 8$	$m_{\text{eff}}^2 = 4.55 g$
$\langle \phi^2 \rangle$ scaling	$\sim \frac{1}{2\pi} \ln L$
Single-kernel skewness	-0.075
Single-kernel kurtosis	2.652
50-term CLT skewness	+0.061
50-term CLT kurtosis	3.012
50-term CLT KS p	0.048
$N/4$ vs mass Spearman	-0.9965
$N/4$ vs mass p -value	4×10^{-12}
$k = 2$ cluster sizes	[10, 4]
$k = 3$ cluster sizes	[8, 4, 2]
SM labeling permutation p	0.045

B Reproducible Verification

The following Python code reproduces the Hodge–Dirac spectrum result.

```
import numpy as np

L = 8; N = L * L

def site_idx(x1, x2, L=L):
    return (x1 % L) * L + (x2 % L)

def edge_idx(x1, x2, mu, L=L):
    return ((x1 % L) * L + (x2 % L)) * 2 + mu

def build_d0(L=L):
    N = L * L
```

```

D0 = np.zeros((2*N, N))
for x1 in range(L):
    for x2 in range(L):
        j = site_idx(x1, x2)
        for mu in range(2):
            i = edge_idx(x1, x2, mu)
            D0[i, j] -= 1.0
            jn = site_idx(x1+1, x2) if mu == 0 else site_idx(x1, x2+1)
            D0[i, jn] += 1.0
    return D0

def build_d1(L=L):
    N = L * L
    D1 = np.zeros((N, 2*N))
    for x1 in range(L):
        for x2 in range(L):
            i = site_idx(x1, x2)
            D1[i, edge_idx(x1, x2, 0)] += 1.0
            D1[i, edge_idx(x1, x2+1, 0)] -= 1.0
            D1[i, edge_idx(x1+1, x2, 1)] += 1.0
            D1[i, edge_idx(x1, x2, 1)] -= 1.0
    return D1

d0 = build_d0(); d1 = build_d1()
assert np.allclose(d1 @ d0, 0)

dim = 64 + 128 + 64
D = np.zeros((dim, dim))
D[ 0: 64,  64:192] = d0.T
D[ 64:192,  0: 64] = d0
D[ 64:192, 192:256] = d1.T
D[192:256,  64:192] = d1

D2 = D @ D
D2_0 = D2[ 0: 64,  0: 64]

Lap = np.zeros((64, 64))
for x1 in range(L):
    for x2 in range(L):
        i = site_idx(x1, x2)
        Lap[i, i] = 4.0
        Lap[i, site_idx(x1+1, x2)] -= 1
        Lap[i, site_idx(x1-1, x2)] -= 1
        Lap[i, site_idx(x1, x2+1)] -= 1
        Lap[i, site_idx(x1, x2-1)] -= 1

assert np.allclose(D2_0, Lap)
print("D^2 == scalar Laplacian on Omega^0: confirmed")

for sl, name in [(slice(0,64), "Omega^0"),
                 (slice(64,192), "Omega^1"),
                 (slice(192,256), "Omega^2")]:
    sub = D2[sl, sl]
    eigs = np.sort(np.linalg.eigvalsh(sub).real)
    u, c = np.unique(np.round(eigs, 6), return_counts=True)
    m4 = c[np.argmax(np.abs(u - 4.0))]
    print(f"{name}: 14-fold at lambda=4 -> mult = {m4}")

```

C Summary of Corrections

Table 5: Corrections to previously circulated WIN papers.

Paper	Correction
SYK_TENSOR_TOE_II	$\gamma = 0.703$, not 1.0–1.2; remove $G_{\text{eff}} = 1/\lambda_L$; remove 4D gravity claim
SYK_TENSOR_TOE	Same
SYK_TENSOR_BLACK_HOLE	Same
SYK_TENSOR_FORCE_UNIFICATION	Remove structure from FDLM/WIN, not SYK
cosmological_constant	Weaken $1/N^2$ claim; remove chaos-gravity mechanism
UV.pdf	$\sigma^2 = 1/2$; $\gamma = -b$
TOE_2_7_2026	$\sigma^2 = 1/2$; $\gamma = -b$
WIN_Paradigm_Explained	Add architecture note; update §11.3, §11.4
The_SYK_Antipode_Theorem	No change
The_SYK_Antipode_Correspondence	No change (superseded)

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